

BASAVARAJESWARI GROUP OF INSTITUTIONS

## BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT



NACC Accredited Institution\*  
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to  
Visvesvaraya Technological University, Belagavi)  
"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,  
Ballari-583 104 (Karnataka) (India)  
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



### DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

## “INDIA AQI DASHBOARD”

A report submitted in partial fulfillment of the requirements for the

### DATA VISUALIZATION WITH POWER BI

Submitted By

NAME: MD BASHEER KHAN

USN: 3BR23CD059

Under the Guidance of  
Mr.Sundeeep Sindhanoor  
Asst. Professor

Dept of CSE (DATA SCIENCE),  
BITM, Ballari



## Visvesvaraya Technological University

Belagavi, Karnataka 2025-2026

BASAVARAJESWARI GROUP OF INSTITUTIONS

**BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT**

NACC Accredited Institution\*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to  
Visvesvaraya Technological University, Belagavi)

"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,  
Ballari-583 104 (Karnataka) (India)

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



**DEPARTMENT OF CSE (DATA SCIENCE)**

**CERTIFICATE**

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"INDIA AQI DASHBOARD"** has been successfully presented by **MD BASHEER KHAN** bearing USN **3BR23CD059** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

**Mr. Sundeep Sindhanoor**

Signature of HOD

**Dr. Aradhana D**

## **ACKNOWLEDGEMENT**

The satisfactions that accompany the successful completion of our mini project on **“INDIA AQI DASHBOARD”** would be incomplete without the mention of people who made it possible, whose noble gesture, affection, guidance, encouragement and support crowned my efforts with success. It is our privilege to express our gratitude and respect to all those who inspired us in the completion of our mini-project. We are extremely grateful to our Guides **Mr.Sundeeep Sindhanoor** for their noble gesture, support co-ordination and valuable suggestions given in completing the mini-project. We also thank **Dr. Aradhana D**, H.O.D. Department of CSE(DS), for his co-ordination and valuable suggestions given in completing the mini-project. We also thank the Principal, Management and non-teaching staff for their co-ordination and valuable suggestions given to us in completing the project.

NAME

USN

MD BASHEER KHAN

3BR23CD059



# India AQI Dashboard using Power BI

Submitted By

Name: Md Basheer Khan

USN: 3BR23CD059

## Table of Contents

| Ch No | Chapter Name                                                                                                                                                        | Page |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1     | Introduction<br><br>1.1 About the project<br><br>1.2 Language used                                                                                                  | 4-5  |
| 2     | Scope of the project<br><br>2.1 Problem statement<br><br>2.2 Objectives                                                                                             | 6    |
| 3     | System requirements<br><br>3.1 Functional Requirements<br><br>3.2 Non-Functional Requirements<br><br>3.3 Hardware Requirements<br><br>3.4 Software Requirements     | 7-8  |
| 4     | Methodology<br><br>4.1 Data Collection<br><br>4.2 Data Preparation(Power Query)<br><br>4.3 Data Visualization<br><br>4.4 Data Modelling<br><br>4.5 Dashboard Design | 9-12 |

|   |                                                                                                                                                                                                                                                                                                     |       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|   | 4.6 outcome                                                                                                                                                                                                                                                                                         |       |
| 5 | Implementation<br><br>5.1 Dataset Description<br><br>5.2 Data Loading<br><br>5.3 Data Transformation(Power Query)<br><br>5.4 Data Modelling<br><br>5.5 DAX Measures(Key Calculations)<br><br>5.6 Dashboard Visualization<br><br>5.7 Dashboard Layout<br><br>5.8 Integration<br><br>5.9 Final output | 13-17 |
| 6 | Result                                                                                                                                                                                                                                                                                              | 18-20 |
|   | Conclusion<br><br>References                                                                                                                                                                                                                                                                        | 21    |

## **ABSTRACT**

The “India AQI Dashboard” is an interactive data visualization project developed using Power BI to analyze air quality across different states in India. The dashboard transforms raw air pollution data into meaningful insights through visual representations such as charts, maps, and KPI indicators. The system allows users to monitor important metrics like Average AQI, Maximum AQI, Minimum AQI, pollutant contributions (PM2.5, PM10, NO2, SO2, CO), and health impact categories. It also provides geographical visualization using maps and trend analysis over time using line charts. Interactive filters such as state, city, and date enable users to explore data dynamically. The dashboard helps in identifying highly polluted regions, understanding pollution trends, and supporting decision-making for environmental monitoring. Overall, this project simplifies complex air pollution data into an easy-to-understand and visually engaging format, making it useful for researchers, policymakers, and the general public.

# CHAPTER 1

## INTRODUCTION

### 1.1 About the Project

Air pollution has emerged as one of the most critical environmental and public health challenges of the 21st century. Rapid urbanization, industrialization, and increasing vehicle emissions have significantly contributed to deteriorating air quality across various regions. Monitoring air pollution levels and understanding their impact is essential for effective environmental management and policy-making. To address this need, the “India AQI Dashboard” has been developed using Microsoft Power BI. This interactive dashboard provides clear, accessible, and data-driven insights into air quality across different states in India. It enables users to analyze pollution patterns, identify highly polluted regions, and understand the impact of various pollutants.

By utilizing a comprehensive dataset containing AQI values, pollutant concentrations (such as PM2.5, PM10, NO2, SO2, and CO), and geographical information, the project transforms complex environmental data into a visually engaging and easy-to-understand format. The dashboard allows users to explore trends over time, compare pollution levels across states, and evaluate health impact categories. Through dynamic visualizations, filtering capabilities, and key performance indicators (KPIs), users can seamlessly analyze air quality data and gain meaningful insights. This project

ultimately bridges the gap between raw environmental data and informed decision-making, serving as a valuable tool for researchers, policymakers, and the general public in understanding and addressing air pollution challenges.

## 1.2 Language Used

The core tools used in this project are:

- **Power BI:** Used for data modeling, formatting, dashboard creation and visualisation.
- **DAX (Data Analysis Expressions):** Used in Power BI to create dynamic calculations, measures, and KPIs such as YoY changes, average carbon intensity, and total emissions.
- **M (Power Query Formula Language):** Used for ETL processes including data cleaning, transformation, handling missing values, and structuring data for analysis.
- **CSV (Data Source Format):** The primary data source format used to import the global energy dataset into Power BI for further processing and visualization

Together, these tools form a cohesive system that efficiently handles two-table relational data modeling, transformation and analysis, ultimately delivering clear, actionable insights through a user-friendly AQI dashboard.

## **CHAPTER 2**

### **SCOPE OF THE PROJECT**

#### **2.1 Problem Statement**

Air pollution data is often large, complex, and difficult to interpret in its raw form. With increasing pollution levels across different states and cities, it becomes challenging for users to analyze trends, compare regions, and understand the impact of various pollutants on human health.

Existing data sources provide numerical values such as AQI and pollutant levels, but they lack proper visualization and interactive analysis. This makes it difficult for researchers, policymakers, and the general public to draw meaningful conclusions from the data.

To address this problem, this project presents an interactive Power BI dashboard that transforms raw air pollution data into clear and visually engaging insights. The dashboard enables users to monitor AQI trends, identify the most polluted states, analyze pollutant contributions, and understand health impact categories. By providing dynamic filtering and visualization capabilities, the system simplifies complex data and supports effective decision-making.

#### **2.2 Objectives**

The main objectives of the project are as follows:

- To analyze air pollution data across different states in India
- To visualize AQI trends over time using interactive charts

- To identify the most polluted states based on AQI values
- To display the contribution of different pollutants such as PM2.5, PM10, NO2, SO2, and CO
- To represent geographical distribution of pollution using map visualization
- To classify air quality into different health impact categories (Good, Moderate, Unhealthy, etc.)

## CHAPTER 3

# SYSTEM REQUIREMENTS

### 3.1 Functional Requirements

- • Import air pollution dataset (2024) from CSV/Excel files and perform data cleaning, transformation, and structuring using Power Query.
- • Provide dynamic filtering and slicing options for state, city, and date to allow interactive data exploration.
- • Calculate and display key metrics using DAX such as Average AQI, Maximum AQI, Minimum AQI, and AQI Alert levels.
- • Analyze pollutant levels including PM2.5, PM10, NO2, SO2, and CO and display their contribution percentages.
- • Enable cross-filtering so that selecting any visual (state, city, or chart) updates all other visuals dynamically.
- • Categorize air quality into health impact levels such as Good, Moderate, Unhealthy, Very Poor, and Severe.

## 3.2 Non-Functional Requirements

- **Performance:** Dashboard interactions, filters, and visuals should load and respond quickly (within ~3–5 seconds) for a smooth user experience.
- **Usability:** The interface should be intuitive, clean, and easy to navigate, with clear labels, readable visuals, and proper color contrast.
- **Scalability:** The data model should support future data additions without significant performance issues.
- **Maintainability:** DAX measures and Power Query (M) steps should be well-structured and documented for easy updates and debugging.
- **Security:** Access to the dashboard and dataset should be restricted to authorized users, especially when deployed on Power BI Service.
- **Reliability:** The dashboard should consistently provide accurate results and stable performance without errors or failures.

## 3.3 Hardware Requirements

- **Operating System:** Windows 10, Windows 11, or Windows Server 2016/2019/2022.
- **Processor (CPU):** 1.4 GHz 64-bit (x64) processor or better.
- **Memory (RAM):** 4 GB RAM.
- **Storage:** 2 GB of available hard disk space (for the Power BI Desktop application and cache).
- **Display:** 1440 x 900 or 1600 x 900 (16:9) resolution.

## 3.4 Software Requirements

- **Development:** Microsoft Power BI Desktop (requires Windows 10/11).
- **Data Source:** CSV files (can be managed via Microsoft Excel or standard text editors).
- **Deployment & Hosting:** Power BI Service (cloud platform for publishing and sharing the

dashboard).

- **End-User Access:** Any modern web browser (Google Chrome, Microsoft Edge, Safari) or the Power BI Mobile App (iOS/Android) for viewing the report.

## CHAPTER 4

### METHODOLOGY

The methodology for the “India AQI Dashboard” focuses on transforming raw air pollution data into meaningful and interactive visual insights using Microsoft Power BI. The project follows a structured workflow that includes data collection, data cleaning and transformation, data modeling, and visualization. Instead of relying on traditional database systems, this project uses CSV/Excel datasets and processes them using Power Query for data preparation. DAX is used for calculating key performance indicators, and Power BI visuals are used to represent the data in an intuitive and interactive manner. This approach ensures efficient handling of data and enables users to explore air pollution trends dynamically.

#### 4.1 Data Collection

- **Dataset Overview:** The project uses a statewise air pollution dataset in CSV/Excel format containing AQI values and pollutant data for the year 2024.

- **Geographical Data:** Includes fields such as State, City, Latitude, and Longitude for map visualization.
- **Temporal Data:** Contains date-wise records to analyze AQI trends over time.
- **Pollutant Data:** Includes values for major pollutants such as PM2.5, PM10, NO2, SO2, and CO.
- **Health Impact Data:** Categorizes air quality into levels such as Good, Moderate, Unhealthy, Very Poor, and Severe.
- **Usage:** The dataset is imported into Power BI and used as the primary source for analysis and visualization.

## **4.2 Data Preparation (Power Query)**

Power Query is used to clean and transform the dataset to ensure accuracy and consistency. The following steps are performed:

- **Handling Missing Values:** Null or missing values are identified and handled appropriately to avoid errors in analysis.
- **Removing Duplicates:** Duplicate records are removed to maintain data integrity.
- **Standardizing Columns:** Column names are simplified and standardized for better readability.
- **Data Type Conversion:** Fields such as Date, AQI, and pollutant values are converted into appropriate data types.
- **Creating New Columns:** Additional columns such as AQI Category and Health Impact are created based on AQI values.
- **Data Structuring:** The dataset is organized in a structured format suitable for analysis in Power BI.

## **4.3 Data Visualization**

The cleaned and processed air pollution data is visualized using various interactive charts and visuals in Power BI. These visualizations help in understanding pollution trends, comparing regions, and analyzing the impact of different pollutants.

Visuals Used:

- Key Performance Indicators (KPI Cards) → Display important metrics such as Average AQI, Maximum AQI, Minimum AQI, Most Polluted State, and AQI Alert level for quick insights.
- Line Chart → Illustrates AQI trends over time, helping users understand how air quality changes across different dates.
- Bar Chart → Highlights the top polluted states based on AQI values, enabling easy comparison between regions.
- Stacked Bar Chart → Shows the contribution of different pollutants such as PM2.5, PM10, NO2, SO2, and CO.
- Pie Chart → Represents the distribution of air quality categories such as Good, Moderate, Unhealthy, Very Poor, and Severe.
- Map Visualization → Displays the geographical distribution of air pollution using latitude and longitude, helping identify pollution hotspots across different locations.
- These visuals are integrated into a single dashboard layout, providing a comprehensive and interactive view of air pollution data.

## 4.4 Data Modeling

The data model is designed to establish relationships between different fields such as Date, Location, and AQI values. A separate date table is created to enable proper time-based analysis.

DAX measures are used to calculate key metrics such as:

- Average AQI
- Maximum AQI
- Minimum AQI
- AQI Alert Levels

The model ensures efficient data processing and supports dynamic filtering across visuals.

## 4.5 Dashboard Design

The final dashboard is designed to present air pollution insights in a clear, structured, and visually appealing layout.

- Top Section: Displays Key Performance Indicators (KPIs) such as Average AQI, Maximum AQI, Minimum AQI, Most Polluted State, and AQI Alert for quick understanding.
- Left Panel: Includes interactive filters/slicers such as State, City, and Date, allowing users to refine and explore the data dynamically.
- Center Area: Shows the AQI trend over time using a line chart, helping users analyze how air quality changes across different dates.
- Bottom Section: Displays comparative visuals such as bar charts for top polluted states and stacked charts for pollutant contribution (PM2.5, PM10, NO2, SO2, CO).
- Right Section: Includes map visualization for geographical distribution of air pollution and a pie chart for air quality (health impact) distribution.

The structured layout ensures easy navigation, better readability, and quick understanding of air pollution patterns.

## 4.6 Outcome

- • Interactive Dashboard: Successfully developed a Power BI dashboard for analyzing statewise air pollution data.
- • AQI Analysis: Enabled clear visualization of Average AQI, Maximum AQI, and Minimum AQI across different regions.
- • Pollution Trends: Provided insights into AQI trends over time using line charts.
- • Pollutant Contribution: Identified the contribution of major pollutants such as PM2.5, PM10, NO2, SO2, and CO.
- • Geographical Insights: Visualized pollution distribution using map representation.

- • Health Impact Analysis: Categorized air quality into Good, Moderate, Unhealthy, Very Poor, and Severe levels.
- • Comparative Analysis: Allowed comparison of pollution levels across different states.
- • Efficient Data Processing: Implemented data cleaning and transformation using Power Query and calculations using DAX.
- • Decision Support: Provided meaningful insights to support environmental awareness and data-driven decisions.

## **CHAPTER 5**

### **IMPLEMENTATION**

The implementation phase of this project focuses on transforming statewise air pollution data into meaningful insights using Microsoft Power BI. The process includes loading the dataset, cleaning and transforming data using Power Query, creating calculations using DAX, and designing an interactive dashboard.

This workflow ensures accurate analysis of air quality trends, pollutant contributions, and health impacts through visually interactive reports.

#### **5.1 Dataset Description**

The dataset used in this project consists of air pollution data across different states and cities in India.

The dataset includes the following key fields:

- State – Name of the state
- City – Name of the city
- Date – Date of observation
- AQI – Air Quality Index value
- Pollutants – PM2.5, PM10, NO2, SO2, CO
- Health Impact – Categories such as Good, Moderate, Unhealthy, Very Poor, Severe

- Latitude & Longitude – Used for geographical visualization

The dataset is imported into Power BI using CSV format.

## 5.2 Data Loading

- The dataset is loaded into Power BI using the “**Get Data**” option.
- Data is previewed and verified before processing.

## 5.3 Data Transformation (Power Query)

- Removed null and duplicate values to improve data quality
- Renamed columns into clear and user-friendly names
- Converted Date column into proper date format
- Converted AQI and pollutant values into numeric data types
- Standardized pollutant fields such as PM2.5, PM10, NO2, SO2, and CO
- Created AQI categories based on air quality levels
- Structured location data including State, City, Latitude, and Longitude
- Prepared data for accurate analysis and visualization

## 5.4 Data Modeling

- **Model Structure:** The project uses a single-table structure where all geographical, temporal, and pollution-related data is stored in one dataset for simplicity and efficiency.
- **Relationships:** A date table is created and linked with the main dataset to enable proper time-based analysis without data inconsistency.
- **Drill-Down Hierarchy:** A hierarchy (State → City) is implemented to allow users to navigate from overall insights to detailed location-level analysis easily.

- **Efficiency:** The model improves performance, enables smooth cross-filtering, and enhances user interaction across dashboard visuals.

## 5.5 DAX Measures (Key Calculations)

To generate insights, the following measures are created:

- **Average AQI:** Calculates the average air quality index across the selected data.
- **Maximum AQI:** Identifies the highest pollution level recorded in the dataset.
- **Minimum AQI:** Identifies the lowest pollution level recorded.
- **AQI Alert:** Categorizes air quality into levels such as Good, Moderate, Unhealthy, Very Poor, and Severe based on AQI values.
- **Most Polluted State:** Determines the state with the highest AQI using ranking logic.
- **AQI Trend:** Tracks changes in AQI over time for trend analysis.

These measures are used in KPI cards and visualizations.

## 5.6 Dashboard Visualizations

Main Visuals

- **KPI Cards:** Display key metrics such as Average AQI, Maximum AQI, Minimum AQI, Most Polluted State, and AQI Alert for quick insights.
- **Line Chart:** Shows AQI trends over time, helping to analyze changes in air quality across different dates.
- **Bar Charts:** Display and compare top polluted states based on AQI values.

- **Stacked Column Charts:** Represent the contribution of different pollutants such as PM2.5, PM10, NO2, SO2, and CO.
- **Pie / Donut Charts:** Illustrate the distribution of air quality levels (Good, Moderate, Unhealthy, Very Poor, Severe).
- **Map Visualization:** Displays geographical distribution of air pollution using latitude and longitude for different cities and states.

## 5.7 Dashboard Layout

The dashboard is organized in a structured manner to ensure seamless navigation and high readability:

- **Top → KPI Cards** (Average AQI, Maximum AQI, Minimum AQI, Most Polluted State, and AQI Alert).
- **Left side → Filters / Slicers** (State, City, and Date) for easy data selection and analysis.
- **Center → Trend Visualizations** (Line chart showing AQI trend over time).
- **Bottom Left → Pollutant Contribution** (Stacked chart showing PM2.5, PM10, NO2, SO2, and CO levels).
- **Bottom Center → Comparison** (Bar chart displaying top polluted states based on AQI).
- **Right side → Geographical & Distribution Analysis** (Map visualization and pie chart showing air quality distribution).

This layout ensures a logical flow, allowing users to move from overall insights to detailed analysis efficiently.

## **5.8 Integration**

- Data is imported from Excel/CSV files containing air pollution data
- All visuals are connected to a single dataset for consistency
- Relationships are created between Date and main data for time analysis
- Filters and slicers dynamically update all visuals
- Map visualization is integrated using latitude and longitude data

## **5.9 Final Output**

The implementation results in a fully interactive Power BI dashboard that:

- Provides real-time insights into air pollution levels across states and cities
- Visualizes AQI trends over time to analyze pollution patterns
- Displays key metrics such as Average AQI, Maximum AQI, and Minimum AQI
- Highlights the most polluted states using comparative bar charts
- Shows pollutant contribution including PM2.5, PM10, NO2, SO2, and CO
- Represents air quality levels through distribution charts
- Enables interactive analysis using filters for State, City, and Date
- Supports geographical analysis through map visualization
- Helps users understand health impact levels based on AQI categories

## CHAPTER 6

### RESULT

The developed “India AQI Dashboard” successfully provides an interactive and visually rich analysis of air quality data across different states and cities. The dashboard transforms raw air pollution data into meaningful insights using various visualizations such as line charts, bar charts, pie charts, maps, and KPI cards.

The trend analysis using line charts clearly shows the variation of AQI over time, helping users understand how air quality changes across different periods. The bar charts effectively highlight the most polluted states, allowing easy comparison of pollution levels across regions.

The pollutant contribution visualization provides a clear understanding of how different pollutants such as PM2.5, PM10, NO2, SO2, and CO contribute to overall air pollution. The pie chart representation helps in identifying the distribution of air quality levels such as Good, Moderate, Unhealthy, Very Poor, and Severe.

Interactive filters and slicers such as State, City, and Date allow users to customize their analysis based on specific requirements. This improves user experience and enables detailed exploration of air pollution data.

The KPI cards display Average AQI, Total Cities, and Total Readings using Power BI's built-in aggregations. Fields were directly dragged into Card visuals and aggregations were set using the dropdown menu — no DAX measures were required.

Overall, the project successfully achieves its objective of providing a simple, interactive, and informative platform for analyzing air pollution data. It helps users identify pollution trends, compare regions, and understand health impacts, making it useful for environmental awareness, research, and decision-making.

## 🇮🇳 India Air Pollution Dashboard | All States

Real-time trends, pollutants & health impact insights

172.83

Avg AQI

20

Min of AQI

400

Max of AQI

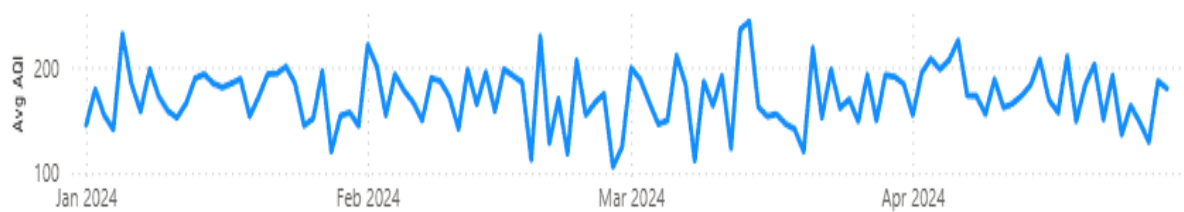
Most Polluted State

Maharashtra

AQI Alert

Normal

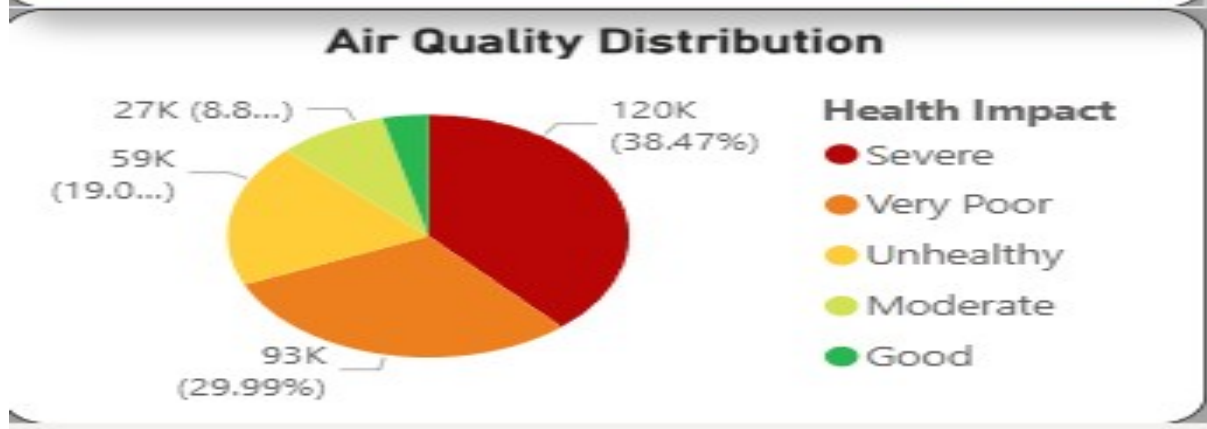
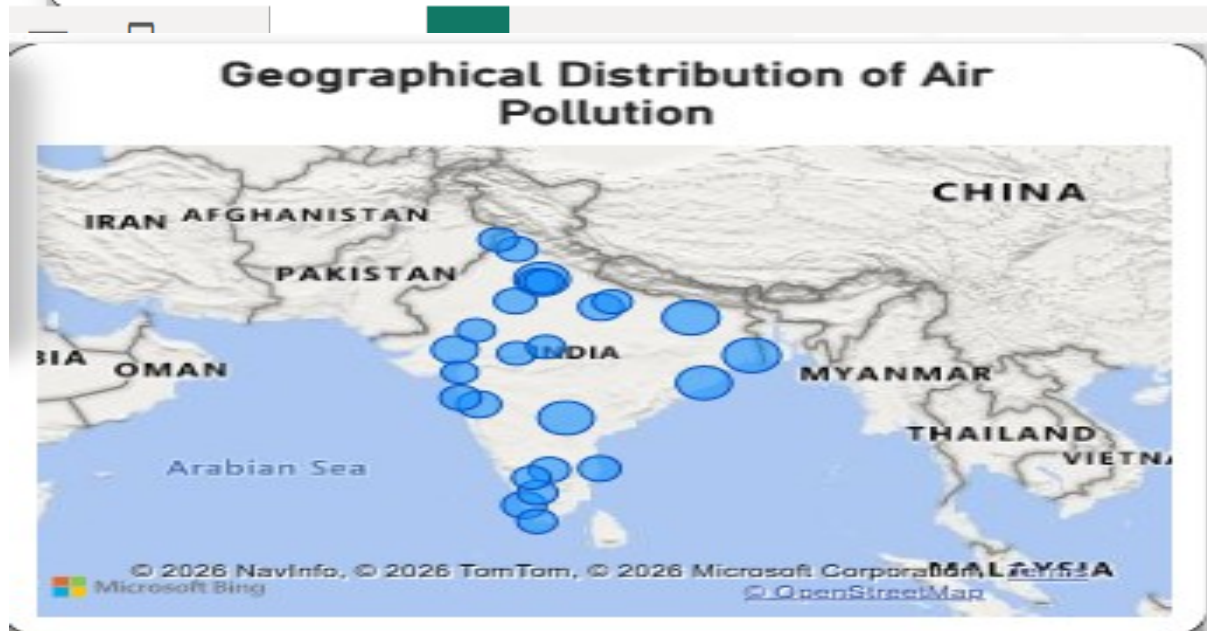
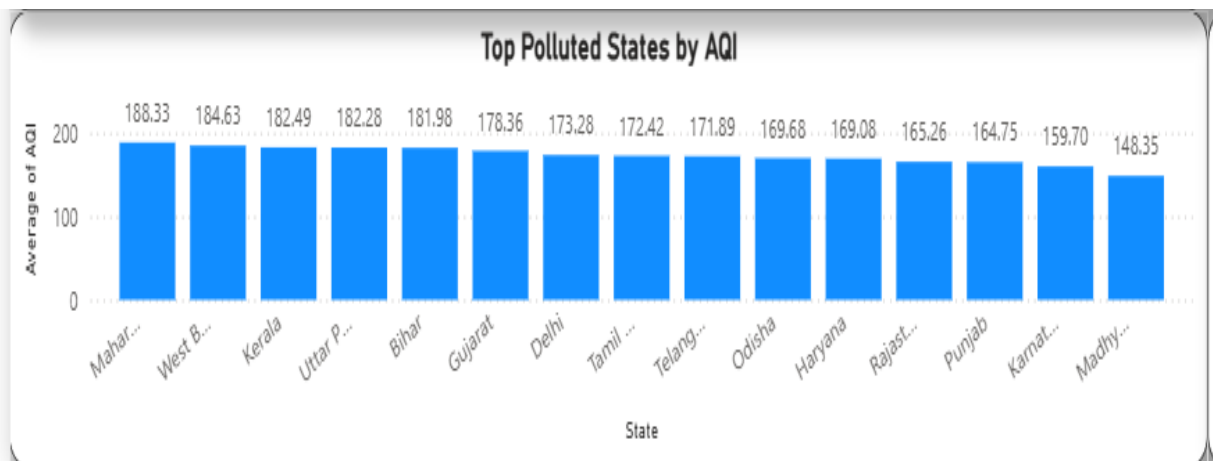
### AQI Trend Over Time



### Pollutant Contribution (%)

CO % NO2 % O3 % SO2 % PM2.5 % PM10 %





## CONCLUSION

The developed “India AQI Dashboard” successfully provides an interactive and visually rich analysis of air quality data across different states and cities. The dashboard transforms raw air pollution data into meaningful insights using various visualizations such as line charts, bar charts, pie charts, maps, and KPI cards.

The trend analysis clearly highlights variations in AQI over time, allowing users to understand changes in air quality across different periods. The bar charts enable easy comparison of pollution levels across states, helping to identify the most polluted regions. The pollutant contribution analysis provides a clear view of how different pollutants such as PM2.5, PM10, NO2, SO2, and CO impact overall air quality.

Interactive filters and slicers such as State, City, and Date allow users to customize their analysis based on specific requirements. This enhances user experience and enables detailed exploration of air pollution data. The KPI cards display key metrics using Power BI’s built-in aggregations by dragging fields directly into visuals, making the project accessible without requiring DAX knowledge.

Overall, the project successfully achieves its objective of providing a simple, interactive, and informative platform for analyzing air pollution data. It helps users identify pollution trends, compare regions, and understand health impacts, making it valuable for environmental awareness, research, and decision-making.

## REFERENCES

- [1] Central Pollution Control Board (CPCB), “National Air Quality Monitoring Programme,” 2024. [Online]. Available: <https://cpcb.nic.in>
- [2] World Health Organization (WHO), “Air Quality Guidelines and Health Impacts,” 2023. [Online]. Available: <https://www.who.int>

[3] IQAir, "World Air Quality Report 2023," IQAir, Switzerland, 2024. [Online]. Available: <https://www.iqair.com/world-air-quality-report>